TB-500, Sometimes Called “TB-500 (Frag)”

Speculative community dose: 2–2.5 mg 2×/wk for 4–6 wk; then 2–2.5 mg weekly

TB-500 is a synthetic N-acetylated heptapeptide, Ac-LKKTETQ, corresponding to residues 17 through 23 of the 43-amino-acid peptide thymosin β4 (Tβ4). Its sequence includes the principal actin-binding motif associated with cytoskeletal regulation and cell migration. Benefits of the acetylated fragment are plausible from related-peptide mechanisms, but the most are for thymosin β4, not TB-500 (unacetylated LKKTETQ). [1,3–9,20] Dermal wound repair, tendon recovery, corneal repair, and broader cardiovascular and neurological tissue repair are investigational applications of TB-500.

💉 Safety: Limited cell-assay and animal observations are reassuring only for the endpoints examined. They do not justify an overall short- or long-term safety rating for injected Ac-LKKTETQ. Immunological, organ, reproductive, and proliferative risks require separate assessment. [3,6,20]

⭐ Efficacy: Skin and tendon repair are plausible, supported by thymosin β4 and incompletely characterized TB-500 experiments. [3–6,20,22]

Cancer caution: Angiogenesis and cell-migration mechanisms create a plausible concern about progression of an existing malignancy. [5,13,20]

Disclaimer: These notes are not authoritative and should not be interpreted as definitive scientific guidance. They are summaries, compilations and plausible extrapolations drawn from the most relevant research I was able to locate, and may omit nuances, conflicting data, or emerging evidence. The material reflects an attempt to organize and understand complex topics, not to provide clinical recommendations or expert interpretation.

This document was updated 6-Nov-26. The most recent version of this document is available at https://pdfs.researcher6076.com/pdfs/tb-500.pdf

Gray Market Confusion Between Tβ4 and TB-500

The names **Tβ4** and **TB-500** are used inconsistently in gray market and research-use-only sales and discussions. In published research, TB-500 is identified as **Ac-LKKTETQ**, a synthetic N-acetylated seven-amino-acid peptide corresponding to residues 17–23 of full-length thymosin beta 4. Full-length Tβ4 contains 43 amino acids. These are distinct molecules, and findings for one should not automatically be attributed to the other. [1,2,20]

Vendor labels may use “TB-500” for either the fragment or full-length Tβ4. Labels such as “TB-500 (TB4)” or “TB-500 (frag)” can help indicate the intended identity, but the name alone does not establish what a sample contains. **Ac-SDKP**, the acetylated four-amino-acid peptide derived from Tβ4’s N-terminus, is another distinct molecule. [1,20,21]

Janoshik reports iuse the following terms. [21]

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Janoshik Terminology	Interpretation
TB-500 (TB-4)	May denote full-length 43-amino-acid Tβ4; verify the actual analyte. [1,20]
TB-500 (Frag)	The cited sample report is consistent with Ac-LKKTETQ, residues 17–23. [1,21]
TB-500 alone	Ambiguous name; consult analytical identity rather than inferring the molecule. [1,20,21]

A COA should be assessed for sample identity, rather than relying on a product name. [20,21]

Dosing Patterns

No evidence-based human subcutaneous dose or interval is established for TB-500. The following dose ranges are speculative experimental patterns, not evidence-based clinical recommendations. [20]

- **2 mg twice weekly for 4 weeks, then 2 mg once weekly during continued recovery.**
- **2.5 mg twice weekly for 4 to 6 weeks, then 2 to 2.5 mg once weekly during continued recovery.**
- **3 mg twice weekly for 6 weeks, then 2.5 mg once weekly during continued recovery. This is not a validated regimen for severe injury or cardiac recovery.**

Dividing the same weekly amount into daily doses is another speculative pattern.

Common Stacking

Combining TB-500 and BPC-157 is a common community practices with plausible benefits. However, in a controlled rat Achilles repair experiment, commercial material labeled TB-500 improved maximum load to failure and histological scores, while combining it with BPC-157 did not add benefit. [6,23]

Benefits

Tβ4 has a broader repair literature. A small rat tendon study found positive outcomes with commercial TB-500 of unconfirmed molecular identity, supporting a plausible repair hypothesis, including possible metabolite-mediated activity, rather than a uniform direct efficacy claim for Ac-LKKTETQ. [3–6,14,20]

- **Dermal wound repair:** Benefits from Ac-LKKTETQ are plausible through migration-related or metabolite-mediated mechanisms. Positive topical aged-mouse findings concern unacetylated LKKTETQ. The parent Ac-LKKTETQ was neutral in the cited fibroblast scratch assay, while Ac-LKKTE improved closure; this endpoint-specific conflict limits a blanket "likely" classification for acetylated TB-500. [3–5,20]

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- **Tendon healing:** Human tendon recovery with Ac-LKKTETQ remains plausible. In a small rodent study daily intraperitoneal commercial TB-500 for 30 days improved maximum load to failure and Bonar/Movin histological scores versus control. [6]
- **Cardiac tissue repair:** TB-500 plausibly supports survival and neovascularization through mechanisms associated with its retained motif. These are mechanistic and related-molecule inferences. [5,10–12]
- **Inflammatory signaling regulation:** Broader anti-inflammatory activity of Ac-LKKTETQ is plausible. A study investigating TB-500 and Ac-SDKP in cell models and 5×FAD mice reported reduced microglial inflammatory polarization, nitric oxide and cytokine production, and glial activation. [7]
- **Neurological tissue repair:** Neuronal-survival and repair hypotheses are plausible. The TB-500/Ac-SDKP investigation reported less neurite atrophy and apoptosis in cell models, greater perirhinal cortical axonal density, and better memory-related performance in 5×FAD mice, without reducing hippocampal amyloid-β burden. Transcriptomic findings implicated apoptosis and synaptic-plasticity pathways, including Foxb2 and Or2k2. [7,18,19]
- **Hair-follicle support:** TB-500 plausibly supports follicular cell migration and remodeling. The cited rodent hair-growth evidence is for full-length Tβ4, involving follicular stem-cell migration/differentiation and matrix metalloproteinase-associated remodeling; it is indirect evidence for Ac-LKKTETQ. [5,15]

Potential Applications

- **Acute soft-tissue repair**. [3–6,14]
- **Chronic overuse injuries**. [5,6,13]
- **Corneal surface repair**. [9,16,17]
- **Post-infarction cardiac repair**. [5,10–12]
- **Neurological recovery**. [7,18,19,22]
- **Hepatic fibrotic signaling**. [8,20]

Cautions

- **Active or previous malignancy**: TB-500's potential effects on angiogenesis and cell migration create a plausible concern that it could support the progression of an existing malignancy. **Active malignancy warrants particular caution**. For a previous malignancy, the relevance of this concern depends on current disease status, recurrence risk, and the biological features of the cancer. [5,13,20]
- **Pregnancy and breastfeeding**: Reproductive, fetal, and lactation safety are unknown. Avoid use during pregnancy and breastfeeding as a precaution. [20]
- **Children and adolescents**: Pediatric safety, dosing, and efficacy are unknown. Avoid use in children and adolescents outside qualified clinical or research oversight. [20]

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Side Effects

Published human exposure is confounded by combinations, and retrospective co-use records cannot establish a TB-500 adverse-effect rate. [20,22,23]

Fatigue, headache, nausea, and transient flu-like symptoms are unverified, anecdotal user reports.

Potential for Diminishing Benefit

TB-500 is not a classical receptor agonist, so receptor desensitization is not an expected primary mechanism of diminished response. [5,13,20]

Biological Mechanism

Full-length thymosin β4 binds globular actin through multiple regions of its sequence and regulates the balance between globular and filamentous actin. This supports cytoskeletal remodeling involved in cell shape, motility, migration, and tissue repair. TB-500 retains the central LKKTETQ actin-binding motif, making effects on actin-related cell migration and repair signaling plausible. The fragment should not be assumed to reproduce the full actin-sequestering activity of thymosin β4. [5,7,8,13,20]

TB-500 and Cancer Risk

TB-500 retains the LKKTETQ region associated with endothelial migration, vessel sprouting, and actin-related cell movement. Because existing tumors can exploit angiogenesis and cell-migration pathways, TB-500 could plausibly support the progression of an existing malignancy. Its effects on cancer initiation and long-term cancer risk are speculative. This mechanistic concern warrants particular caution in people with active malignancy and consideration in those with previous malignancy or elevated recurrence risk. [5,13,20]

Conclusions

While full length thymosin β4 is much studied, TB-500 evidence is mostly inferred from the LKKTETQ region of the full-length peptide. Its applications and benefits are mechanistically plausible. Much of the peptide sold in gray market labeled TB-500 is actually full length thymosin β4, therefore much of the TB-500 discussions about TB-500 are actually discussions about full length thymosin β4.

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Gray-Market Analytical Example

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